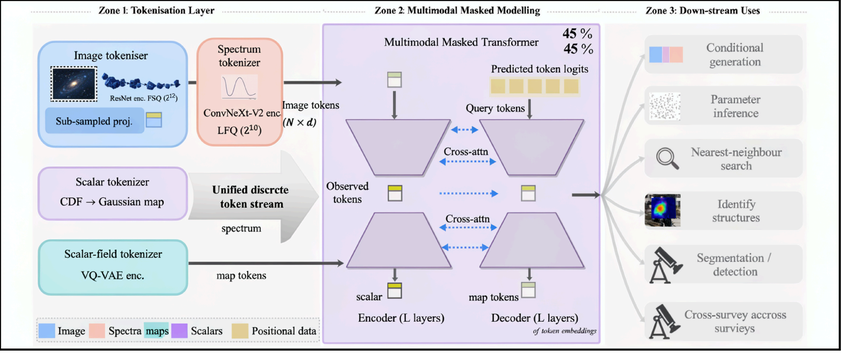
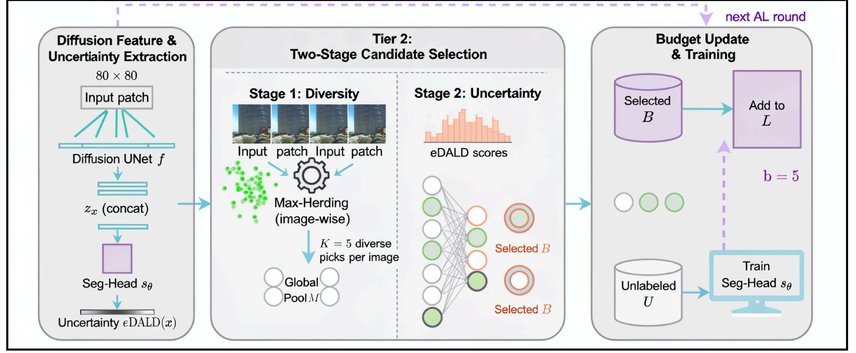
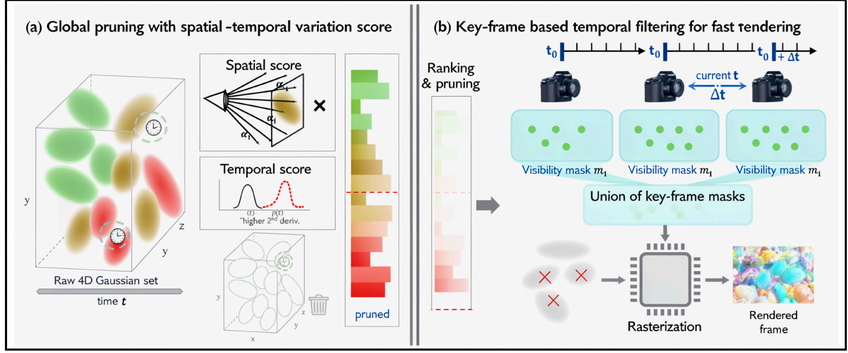
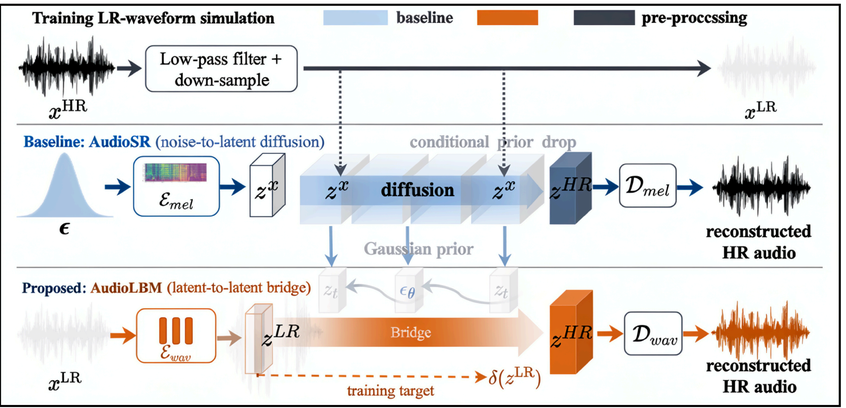
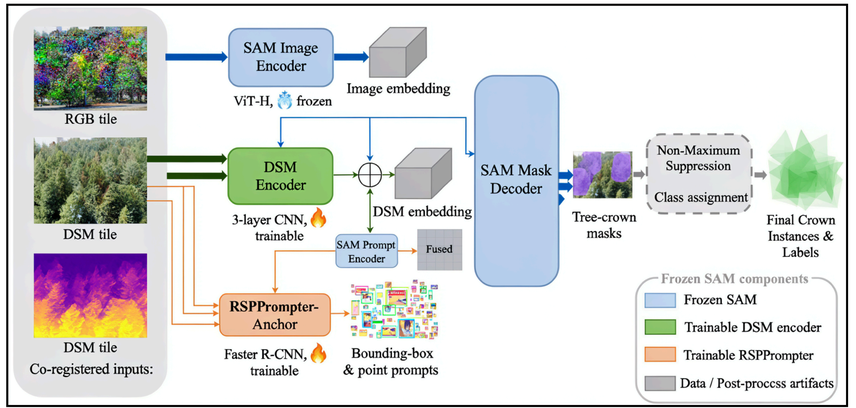
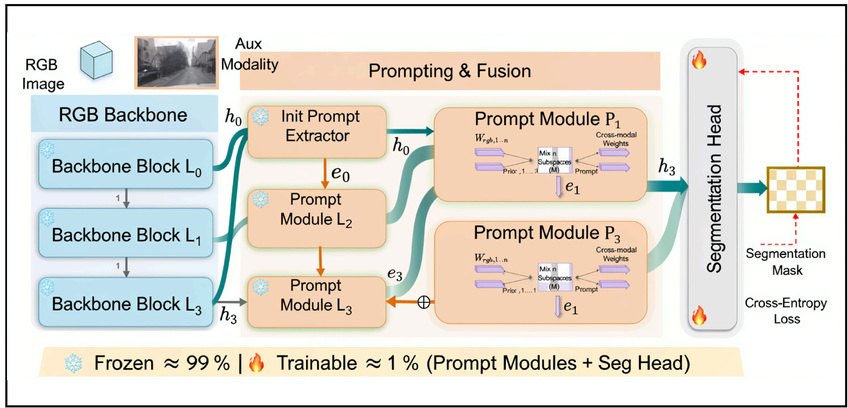

Zone 1: Tokenisation Layer
Zone 2: Multimodal Masked Modelling
Zone 3: Down-stream Uses
Image tokeniser
ResNet enc. FSQ ($2^{15}$)
Sub-sampled proj.
Spectrum tokenizer
ConvNeXt-V2 enc.
LFQ ($2^{10}$)
Image tokens ($N \times d$)
Scalar tokenizer
CDF → Gaussian map
Unified discrete token stream
spectrum
Scalar-field tokenizer
VQ-VAE enc.
map tokens
Multimodal Masked Transformer
45 %
45 %
Predicted token logits
Query tokens
Cross-attn
Observed tokens
Cross-attn
scalar
map tokens
Encoder (L layers)
Decoder (L layers)
of token embeddings
Conditional generation
Parameter inference
Nearest-neighbour search
Identify structures
Segmentation / detection
Cross-survey accross surveys
Image
Spectra maps
Scalars
Positional data


next AL round
Diffusion Feature & Uncertainty Extraction
80 × 80
Input patch
Diffusion UNet $f$
$z_x$ (concat)
Seg-Head $s_\theta$
Uncertainty eDALD($x$)
Tier 2: Two-Stage Candidate Selection
Stage 1: Diversity
Input patch Input patch
Max-Herding (image-wise)
$K = 5$ diverse picks per image
Global Pool $M$
Stage 2: Uncertainty
eDALD scores
Selected $B$
Selected $B$
Budget Update & Training
Selected $B$
Add to $L$
b = 5
Unlabeled $U$
Train Seg-Head $s_\theta$


(a) Global pruning with spatial -temporal variation score
Spatial score
Temporal score
"higher 2nd deriv."
Raw 4D Gaussian set
time $t$
pruned
(b) Key-frame based temporal filtering for fast tendering
Ranking & pruning
current t
Visibility mask $m_1$
Visibility mask $m_1$
Visibility mask $m_1$
Union of key-frame masks
Rasterization
Rendered frame


Training LR-waveform simulation
baseline
pre-proccssing
Low-pass filter + down-sample
$x^{HR}$
$x^{LR}$
Baseline: AudioSR (noise-to-latent diffusion)
conditional prior drop
$\mathcal{E}_{mel}$
$z^x$
diffusion
$z^{HR}$
$\mathcal{D}_{mel}$
$\epsilon$
reconstructed HR audio
Gaussian prior
Proposed: AudioLBM (latent-to-latent bridge)
$z_t$
$\epsilon_\theta$
$\mathcal{E}_{wav}$
$z^{LR}$
Bridge
$\mathcal{D}_{wav}$
training target
$\delta(z^{LR})$


RGB tile
DSM tile
DSM tile
Co-registered inputs:
SAM Image Encoder
ViT-H, frozen
Image embedding
DSM Encoder
3-layer CNN, trainable
DSM embedding
SAM Prompt Encoder
Fused
RSPPrompter-Anchor
Faster R-CNN, trainable
Bounding-box & point prompts
SAM Mask Decoder
Tree-crown masks
Non-Maximum Suppression
Class assignment
Final Crown Instances & Labels
Frozen SAM components
Frozen SAM
Trainable DSM encoder
Trainable RSPPrompter
Data / Post-proccss artifacts


RGB Image
Aux Modality
Prompting & Fusion
RGB Backbone
Backbone Block $L_0$
Backbone Block $L_1$
Backbone Block $L_3$
Init Prompt Extractor
Prompt Module $L_2$
Prompt Module $L_3$
Prompt Module $P_1$
Prompt Module $P_3$
Segmentation Head
Segmentation Mask
Cross-Entropy Loss
Frozen ≈ 99 % | Trainable ≈ 1 % (Prompt Modules + Seg Head)